17. Windows OS Commands Execution (traceroute, ping, ifconfig, netstat)

1.ping command

```
C:\Users\SAIL>ping google.com

Pinging google.com [142.251.221.142] with 32 bytes of data:
Reply from 142.251.221.142: bytes=32 time=3ms TTL=117
Reply from 142.251.221.142: bytes=32 time=3ms TTL=117
Reply from 142.251.221.142: bytes=32 time=4ms TTL=117
Reply from 142.251.221.142: bytes=32 time=4ms TTL=117

Ping statistics for 142.251.221.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 4ms, Average = 3ms
```

2. Traceroute

```
C:\Users\SAIL>tracert google.com

Tracing route to google.com [142.251.221.142]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  172.23.16.1
  2    <1 ms    <1 ms    <1 ms  172.22.8.1
  3     2 ms    <1 ms     1 ms  14.139.187.225
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6     *        *        *     Request timed out.
  7     *        *        *     Request timed out.
  8     3 ms     4 ms     2 ms  lcmaaa-as-in-f14.1e100.net [142.251.221.142]

Trace complete.
```

3.ipconfig

```
C:\Users\SAIL>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::7088:5fd8:76b2:8e47%17
   IPv4 Address. . . . . . . . . . . : 172.23.161.107
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . : 172.23.16.1

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::92e:6818:33d5:35b8%5
   IPv4 Address. . . . . . . . . . . : 192.168.163.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::5209:ba7a:ed11:6a0b%15
   IPv4 Address. . . . . . . . . . . : 192.168.160.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

4.netstat

```
C:\Users\SAIL>netstat -an

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:902            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:912            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:2343           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:3389           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:3580           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:3582           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:8080           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49666          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49667          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49668          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49698          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49703          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49712          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:59110          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:59111          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:59112          0.0.0.0:0              LISTENING
  TCP    127.0.0.1:3580         127.0.0.1:57923        TIME_WAIT
  TCP    127.0.0.1:3580         127.0.0.1:57925        TIME_WAIT
  TCP    127.0.0.1:3580         127.0.0.1:57927        TIME_WAIT
  TCP    127.0.0.1:49671        127.0.0.1:49672        ESTABLISHED
```